Games and Simulations – Test 1

2024-2025

Date: 07/05/2025 Duration:1h15

(VERSION **A**)

Number: _____ Name: _____

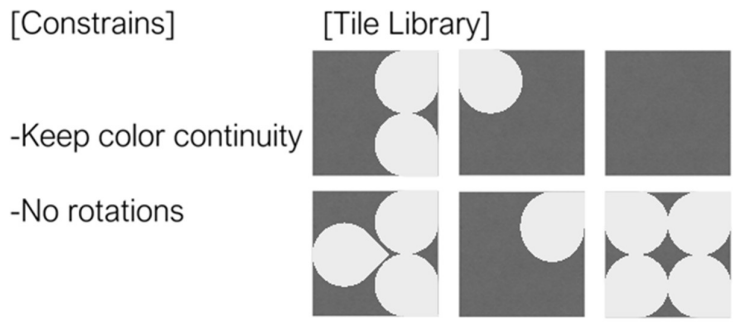
- You may answer in Portuguese or English.
- Do not separate the sheets. You may use a pencil. You may use the back as a draft.
- You can bring a handwritten A4 sheet for consultation and use the back as a draft.
- The test has two groups: Group 1 – answer in the sheet, Group 2 – answer in Akindi.
- Each wrong answer in group 2 has a 20% discount. If you don't answer it's worth 0.

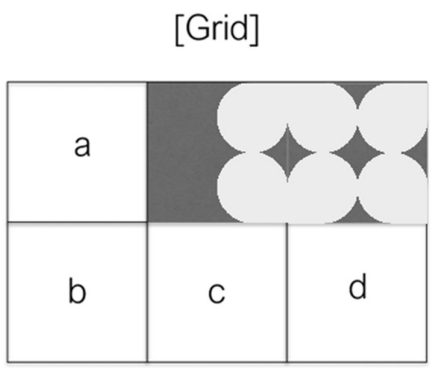
Group 1 – 6 points

1. P. Milgram and A. F. Kishino presented the concept of Mixed Reality in their 1994 paper. Elaborate and explain in detail what Milgram proposed, how does Augmented Reality (AR) and Virtual Reality (VR) fit into his model and what are the main characteristics and differences of AR and VR.

2. You have an indoor game and decided to paint the floor of the house using a pattern that is procedurally generated using a Wave Function Collapse algorithm.

   Considering the following types of tiles in the library, you want to keep color continuity between tiles and you are not allowed to rotate the tiles. Considering that the grid has already two tiles. What are the values of a, b, c and d and which tile will be filled next?





a =

b =

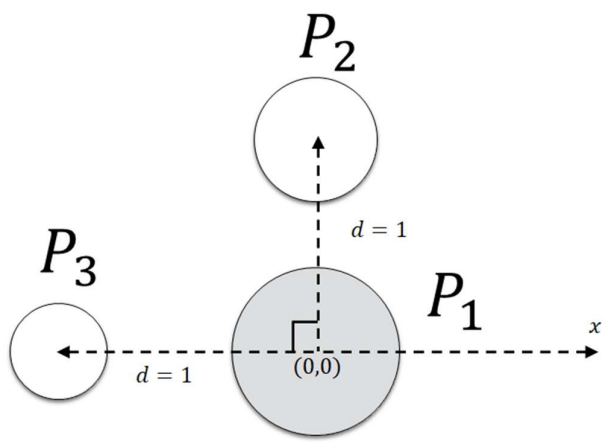
c =

d =

next tile? (a,b,c or d) =

Group 2 – 14 points (answer in Bubble sheet)

[1-3] Consider the following particle system at a certain time *t* in a cartesian system (positive *y* is up, positive *x* is right). The distance of P2 and P3 to the first particle is 1. The system uses a Hooke force model with an equilibrium of 2 and elasticity of 1 (eq = 2, k = 1).
Consider a simulation step ***dt = 3.***



$$S = \begin{bmatrix} (0,0) & ? & ? \\ (0,0) & (3,2) & (0,0) \\ f1 & f2 & f3 \\ 3 & 2 & 1 \end{bmatrix}$$

1. What is the value of ***f1*** in the next step?

   A. f1=(1,-2) B. f1=(-3,3) C. f1=(-3,6) D. f1=(1,-1) E. None of the others

2. Using an Euler Interpolation, what will be the new values of ***v*** of the first particle?

   A. v1=(1,-1) B. v1=(-3,6) C. v1=(4,6) D. v1=(-2,1) E. None of the others

3. Using an Euler Interpolation, what will be the new values of ***p*** of the first particle?

   A. p1=(-1,2) B. p1=(-2,4) C. p1=(-9,18) D. p1=(-3,3) E. None of the others

4. Consider a particle system that follows the hard spheres model in a XY space. Two particles collide with the collision axis ***r*** parallel to the Y axis. Consider that the time before the collision was t, and that the velocities before the collision where:
   v_i = (2, -3)
   v_j = (4, 2)
   σ= 3
   What is the value of the velocity components at t+1:

   A. vi = (4, -3) vj = (2, 2)
   B. vi = (-4, -2) vj = (2,-3)
   C. vi = (2, 2) vj = (4,-3)
   D. vi = (2, 3) vj = (4,-2)
   E. None of the others

5. Consider two particles that are at a certain distance d (||d|| = 2) and have equal mass. The system was tested using two force models.

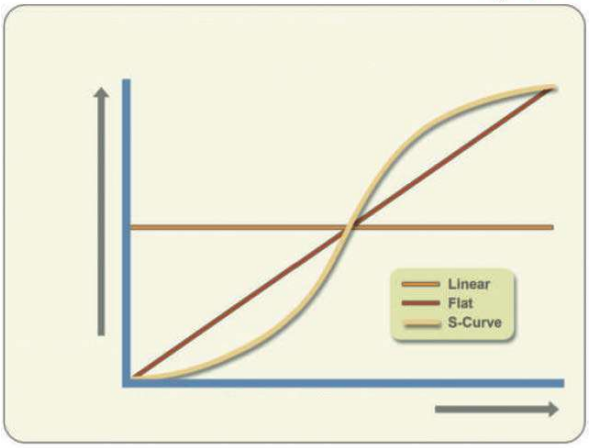
   Test 1: Gravity Model
   Test 2: Hooke model with eq = 2

   Choose the correct answer:
   A. Test 1: the particles repel each other. Test 2: particles repel each other.
   B. Test 1: particles repel each other with low intensity. Test 2: Stationary.
   C. Test 1: particles attract each other with high intensity. Test 2: Stationary.
   D. Test 1: Stationary. Test 2: particles repel each other.
   E. None of the above

6. The s-curve progression can be applied to:

   A. The Flow level.
   B. The Narrative arc.
   C. The Completion Time.
   D. Difficulty of the game mechanics.
   E. None of the others.

Diagram by Per Olin



7. The game designer LeBlanc proposed a taxonomy of ...

   A. Game Pleasures such as Sensation, Space, Fellowship, Actions, Challenge etc.
   B. Player Types such as Achievers, Explorers, Socializers, Killers.
   C. the Eight Filters.
   D. Game Concepts.
   E. None of the other options.

8. Photogrammetry is the use of photography in surveying and mapping to ascertain measurements between objects. Which sentence is correct?

   A. Photogrammetry uses several images to automatically generate different parts of a game level.
   B. Using Photogrammetry reduces the game development time but requires dealing with mesh detection artifacts and defects.
   C. Photogrammetry is a technique used in games to generate dynamic games.
   D. Photogrammetry uses several photos to increase the quality of textures by correctly measuring the size of an object.
   E. None of the others

9. Considering the differences between games and simulations, which sentence is correct?

   A. A game responds to user input, thus is an interactive simulation.
   B. Games are mostly soft, real-time, interactive, agent-based applications, and for that reason are not simulations.
   C. A game is a complete set of a real or imaginary world modeled mathematically and simulated in a computer.
   D. Simulations must run at least at 1/30 of a second.
   E. None of the others.

10. To which game element from the tetrad does the sentence belong:
    1. Sounds that we hear during the explosions.
    2. The main character must use a kite to fly above the river to complete the level.
    3. The main character is in love with the princess from the other kingdom.

A.	B.	C.	D.	E.
1 – Aesthetics 2 – Players 3 – World	1 – Player 2 – Mechanics 3 – Story	1 – Story 2 – World 3 – Player	1 – Aesthetics 2 – Mechanics 3 – Story	None of the others

11. According to Jessel Schell's Taxonomy, Space is one of the game mechanics. Consider the game of checkers (Damas) where the players take turns in pushing pieces in any free diagonal direction. The pieces can only move in the black squares.

    What sentence better describes one of the mechanics of the game:

    A. The space is continuous, 2 dimensions, 8x8 bounded spaces.
    B. None of the others
    C. The space is continuous, 2 dimensions, 4x8 bounded spaces.
    D. The written rules state that you can only move diagonally.
    E. The Player can perform the Operative Action "move vertically forward" if the next square is free.

12. Many modern online games allow the player to buy tokens which can then be used to purchase better weapons, reduce waiting times etc. This creates a revenue stream for the game developers but introduces problems in ...

    A. .. choice, the reward for the player is lower.
    B. none of the others
    C. ... balance, the game becomes asymmetrical.
    D. ... balance, the game becomes symmetrical.
    E. ... choice, the risk for the player is lower.

13. Which type of time is more adequate:
- "The player can pause time in the middle of the game and manually move the bullets away"

A. Isometric B. Authentic C. Limited D. Altered E. None of the others

14. This is a plant generated using L-Systems with the following variables:

angle = 25 , axiom = F , iteration = 2

What is the rule used to create this plant?

A. F = F[+F][-F]FF
B. F = F[+F]F[-F]F
C. F = F+F-FFF
D. F = F+F- -F[F]F
E. F = F[+F][-F]

15. Consider the Game Townscaper represented in the following figure. Which PCG algorithms would be better to create this kind of system:

A. Noise Functions
B. Puzzle Generation, Level Generation
C. Constructive.
D. Voronoi, Wave Function Collapse
E. Delaunay, Marching Cubes

16. Consider a squared texture with a side of 256 pixels, color mode RGB and a color depth capable of representing at least 256 colors in each channel (R,G and B). How much space in bytes does it occupy in memory?

A. 256x256x3x1
B. 256x256x3x256
C. 2x256x3x2
D. None of the others.
E. 2x256x4x16

17. We want to record a sound using a single elementary microphone. The sound has a maximum frequency of 11kHz. The bits per sample are the same as in CD audio.

    Consider the Nyquist–Shannon sampling theorem that is used in the sampling of audio signals. What should be the minimum quality required to sample this sound?

    A. 22kHz, 8bits, Mono
    B. 44kHz, 16bits, Mono
    C. 11kHz, 16bits, Stereo
    D. 22kHz, 16bits, Mono
    E. None of the others

18. Pixel Art is a form of creating 2D assets that tries to recreate the look and feel of old generation 8-bit gaming consoles. Its main characteristic is:
    A. Using pixels to create 2D game characters and objects.
    B. Using a maximum of 256 colors and low resolution.
    C. Using a Japanese Anime drawing style.
    D. Stacking all the images and animations into a single large texture that is divided into sprites.
    E. None of the others.

(to be use as draft)